Circular Linked List

C Program Implementation and Execution Output

PROGRAM SOURCE CODE

```
#include <stdio.h>
#include <malloc.h>

struct node
{
    int data;
    struct node *next;
};

struct node *head, *tail, *newnode, *temp, *prev;

void createnode(int d)
{
    newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = d;
    newnode->next = NULL;
}

void insert(int key, int data)
{
    int found = 0;
    temp = head;
    do
    {
        if(temp->data == key)
        {
            found = 1;
            break;
        }
        temp = temp->next;
    } while(temp != head);

    if(found == 0)
    {
        printf("key element not found \n");
    }
    else
    {
        createnode(data);
        if(temp == tail)
        {
            temp->next = newnode;
            tail = newnode;
            tail->next = head;
        }
        else
        {
            newnode->next = temp->next;
            temp->next = newnode;
        }
        printf("inserted :%d after :%d\n", data, key);
    }
}
```

PROGRAM OUTPUT

```
data appended :10
data appended :20
data appended :30
data prepended :40
40 ->10 ->20 ->30 ->
--
inserted :25 after :20
40 ->10 ->20 ->25 ->30 ->
--
data deleted :10
40 ->20 ->25 ->30 ->
40 ->20 ->25 ->30 ->
```